7(a) In an electron microscope, electrons are accelerated by an electric field in order to investigate the structure of matter. This is done by analyzing the diffraction pattern produced when the accelerated electron beam is passed through the arrangement of atoms in solids. This technique is possible due to the phenomenon known as wave–particle duality.

(i) Explain what is meant by the term *wave–particle duality* as applied to this technique. [1]

(ii) Estimate the kinetic energy of electrons such that they can be used to measure the distance between adjacent atoms in crystals, which is typically about 0.1 nm. [4]

(iii) The percentage uncertainty in measuring the electron's velocity is 10%. Calculate the minimum uncertainty in a simultaneous measurement of the electron's position. [2]

(iv) Calculate the energy of a photon which has the same wavelength as the electron. [2]

(b) Scientists suggests the use of *protons* to replace the electrons in the electron microscope in part (a).

(i) Determine the associated wavelength of protons with the same kinetic energy as the electrons in part (a). [1]

(ii) State and explain whether the use of protons would make a better or less effective probe of small-scale structures. [1]